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UNIVERSITI TEKNOLOGI MALAYSIA

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UTM Johor Bahru

SECP 3204: Software Engineering

Semester 01, 2024/2025

PROJECT PROPOSAL

KADA SYSTEM

LEMBAGA KEMAJUAN PERTANIAN KEMUBU(KADA)

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1. Introduction

Efficient document management system (DMS) is essential for every organization since it significantly reduce the time required for document handling and optimizing organizational workflows. However, Lembaga Kemajuan Pertanian Kemubu (KADA) still rely on traditional physical document management systems which makes it difficult for them to keep up with current needs. KADA currently aims to improve service delivery and optimize internal processes, there is a pressing need to transition from traditional methods to digital solutions that offer greater efficiency, security, and accessibility.

1.1. Need (N)

As highlighted by KADA's representative Mr. Ahmad Rohailan Bin Hani, the current reliance on paper-based document management at KADA results in inefficient workflows, slow processing and increased risks of document loss. Furthermore, the lack of robust reporting features limits data-driven decision-making which making it harder for them to adapt to rapid changes in information management. There is a critical need for an efficient DMS system that can overcome these limitations, streamline document handling and improve operational efficiency. The new KADA system aims to address these needs by digitizing and automating KADA's document workflows, significantly enhancing operational efficiency.

1.2. Approach (A)

To address these challenges, the proposed new KADA system is a cloud-based document management solution tailored for KADA's specific needs. This system will provide a web-based platform that can be accessed on various devices, enabling users to upload, organize, and retrieve document digitally. Key features include an intuitive interface, automated document categorization, advanced search functions, and comprehensive reporting tools to support decision-making.

1.3. Benefit (B)

The new proposed system will significantly improve KADA's document management efficiency by reducing manual handling, accelerating document processing, and minimizing the risk of document loss. Users will benefit from streamlined workflows and have a quicker access to documents. Additionally, the

system's reporting features will enable KADA to generate a detailed reports for better decision-making that leads to a better productivity and service delivery.

1.4. Competitor (C)

Several well-known document management systems, such as Google Drive[1], Microsoft SharePoint[2], and Dropbox[3] offer clouds storage and file-sharing features. However, these platforms do not address the specific needs of government agencies like KADA, particularly in managing member registration, loan applications, and reporting processes.

Google Drive provides secure file storage and sharing but it lacks built-in feature for automating process like report generation. Similarly, Microsoft SharePoint offers some customization options but can be complex to use. Dropbox is ideal for file storage but does not provide tailored solutions for workflow automation or compliance tracking.

The proposed system is specifically designed to address these gaps. It includes advanced reporting features that generate real-time insights for strategic planning. Moreover, the new proposed system ensures compliance tracking to meet KADA standards, reducing associated with audits and documentation errors. It prioritizes a user-friendly interface design and specifically engineered for smooth compatibility to integrate well with KADA's existing IT infrastructure.

2. Existing Systems

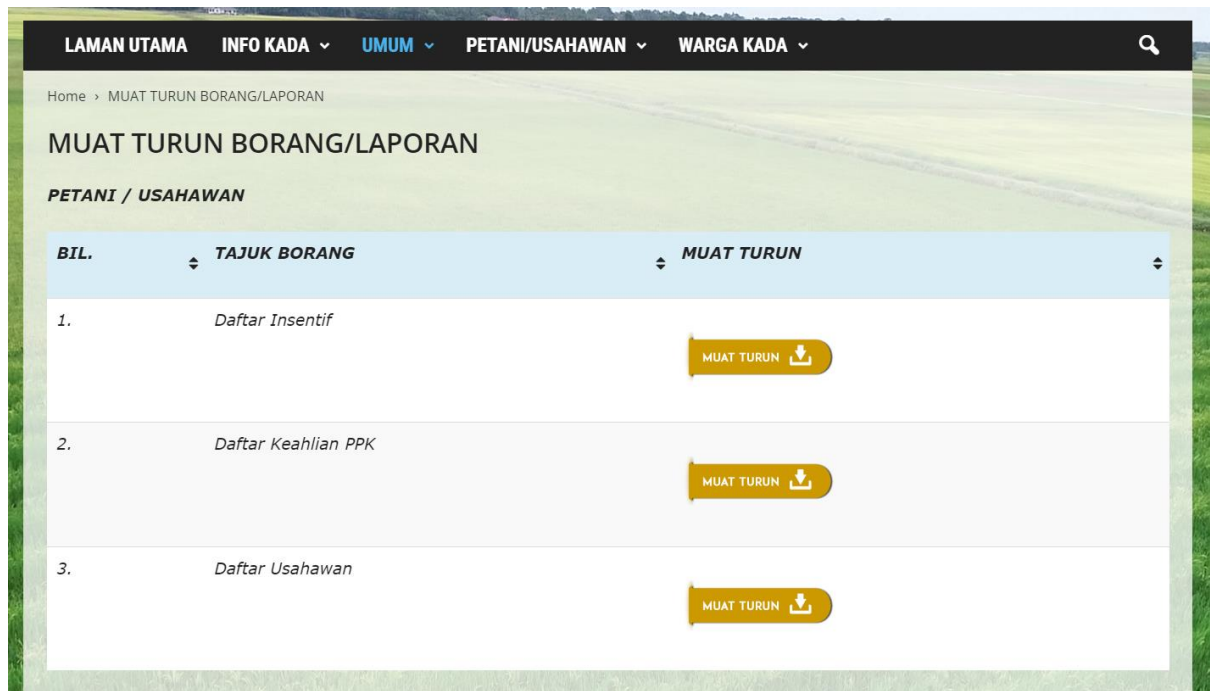


Figure 1 : <http://www.kada.gov.my>

Based on Figure 1 , we can identify that neither the process of member registration nor loan application can be carried out through this website. Users are required to complete manual forms by downloading the forms.

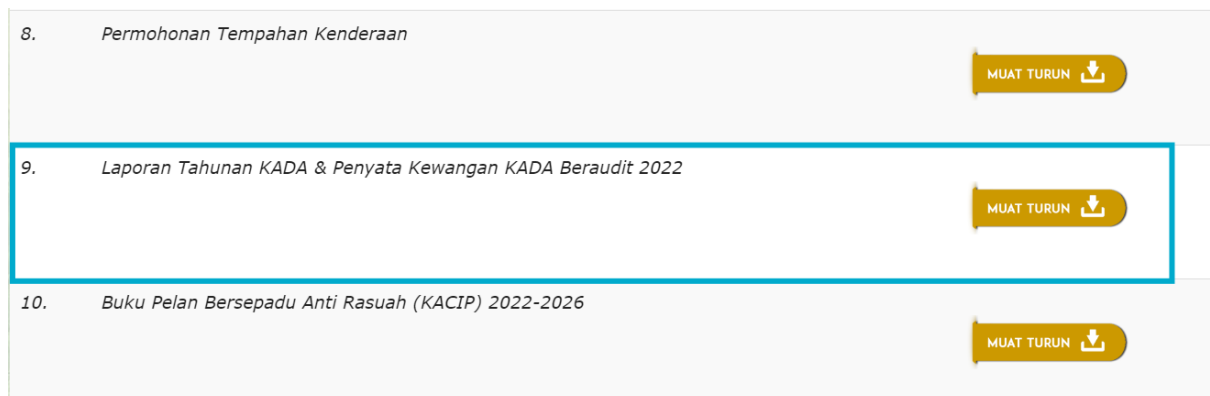


Figure 2 : <http://www.kada.gov.my>

Based on Figure 2, we can identify that the existing system does not support automated generation of annual reports. KADA employees are required to manually enter the data to produce an annual report.

Features	KWSP i-Akaun	Maybank2U	KADA
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Digitised Member Registration	Yes	Yes	No
User Authentication for member login	Yes	Yes	No
Automatic generation of annual report	Yes	Yes	No

Table 1: Comparison of existing systems

3. Proposed System

3.1 Digitise the member registration and data entry process

Changing the paper-based system for member registration and loan application process to a digitise system using website. No more paper forms filling process for member registration and loan application and KADA staff are not required to perform manual data entry to register new members. This will mitigate the risk of using paper forms, ensure data security, saving times for checking the documents and most importantly avoid human error.

3.2 Security

3.2.1 User authentication mechanism

Login page will be created for the system due to security and privacy concerns. After registering an account, the members can login by entering their username and password. Then, they can apply for a loan or check the status of loan applications and payments made monthly. The system is also able to detect users that are not registered as a member to ensure the security of the system.

3.2.2 OTP Verification

OTP verification is a must for verifying the provided phone number is from the person who register as a member. The OTP verification will be used in member registration process and during change of password for verifying the identification of the user. This is a way to secure our website and data privacy of our users.

3.2.3 Probably destroy section

The current feature is to ensure the website will be logged out while the website is stopped using by our user. This feature is to protect the data privacy of our users and strengthen the security of our website.

3.3 Automated Generation of Report

Currently, KADA is still required to generate reports for the board of directors manually. However, by using our system, the report for each member will be automatically generated after the clerk has updated the loan and dividend details in the system. Members can also obtain a copy of their individual report via the homepage.

3.4 Email Notification to clerk

Notification by email will be used to notify the clerk to proceed the steps for checking all the system applications and deciding the result of the application. When there is a new application sent into the system, the system will auto generate an email notifying the clerk to ensure each application will not be ignored.

3.5 User-friendly layout

3.5.1 Customization layout

By consideration, our website can adjust the text size that is suitable for different users as each user have different requirement while using website. Otherwise, there are choices for day or night mode for each user based on their preference.

3.5.2 FAQ

Frequently Asked Question section will be provided to solve issues or problems that our user faces. This section is needed for them as a guidance while facing problems that are commonly faced by most of the people and to save the time for the server's admin for assisting each issue.

3.5.3 Chatbot

In our system, we will provide a live chatbot that simulates human conversation to help users answer questions or solve problems. This feature is more user-friendly as it can solve the user's problem in a short time.

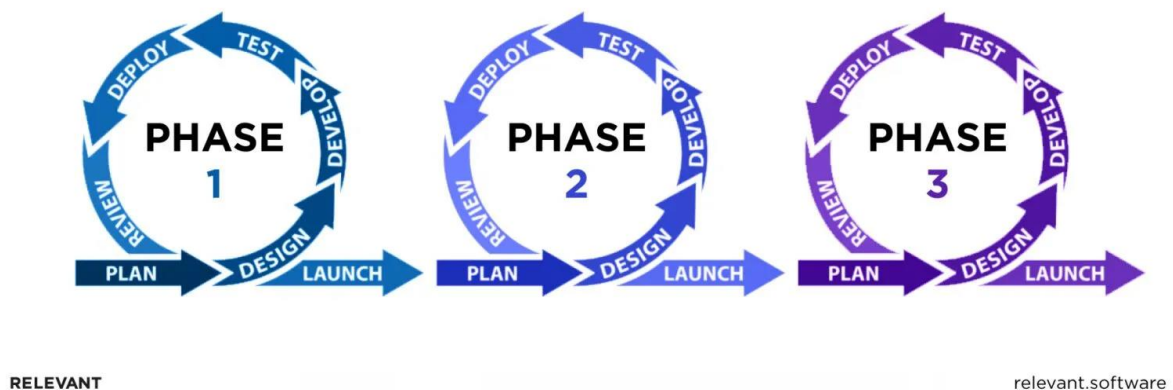
4. Software Process Modal

4.1 Software Process Model – Proposed System Development

The ideal software process model for internally developing our proposed system is the Agile Development Model, chosen for its flexibility, iterative nature, and focus on user-centric outcomes. Agile emphasizes incremental delivery, allowing delivery of the system in small and manageable parts, and each increment represents a usable portion of the system that is created, tested, and deployed iteratively. It enables early feedback and usability testing as it allows features to be released progressively. For instance, the system could be developed into different functional modules, and referred to as increments such as user registration, member login functionalities, loan applications, and so on.

4.2 Software Process – Detailed Activities

In agile software development, the activities involved in each phase are not limited to a linear end-to-end process. Unlike traditional development methods, where each phase occurs sequentially, Agile emphasizes continuous, iterative cycles (sprints) in which activities overlap and evolve as the project progresses. Each Sprint consists of repeated processes such as planning, design, development, testing, deployment, and review, all of which are revisited and improved at the beginning of each iteration. This approach ensures that feedback is continuously incorporated, features are delivered incrementally, and the product can adapt to changing user needs. The following sections outline the detailed activities performed in each Sprint:



1. Planning Phase

- Product Backlog Creation: The team works with stakeholders to create an initial product backlog, which is a prioritized list of features, user stories, and tasks that need to be done for the product.
- Sprint Planning: At the beginning of each Sprint, the team decides which items from the backlog will be worked on during the Sprint.
- Task Breakdown: Each user story or feature is broken down into manageable tasks, and the time estimates are assigned.

2. Design Phase

- Collaboration with Stakeholders: Continuous input from stakeholders helps shape the design according to emerging requirements.
- System Layout: In the beginning, the design is more high-level and conceptual. As iterations go on, the design evolves.

- Refinement: Each iteration may require improvements to the design as new features are added or adjustments made based on feedback. The team adapts the design to the changing needs of the project.

3. Development Phase

- Coding: The developer starts implementing the tasks defined during the Sprint schedule.
- Daily Stand-ups: The team holds a daily stand-up meeting (or daily Scrum) to discuss progress, obstacles, and next steps.
- Continuous Integration: Developers are constantly integrating their work into the main codebase, ensuring that the code works harmoniously with the contributions of others.

4. Testing Phase

- Continuous Testing: As features are developed, they are continuously tested.
- User Acceptance Testing: UAT can take place at the end of each Sprint, especially if the Sprint delivers a functional feature that can be verified with the product owner or stakeholder.

5. Deployment Phase

- Incremental Release: Each Sprint ends with an incremental release of the software, which is a potentially deployable release.
- Deployment: If the product increment is stable and meets acceptance criteria, then it can be deployed or provided for user feedback. When all system layouts are developed and stable, a final round of testing will take place in a production environment.

6. Review/Maintenance Phase

- Bug Fixing: Any bugs found in the current iteration (or previous iterations) will be fixed in the next iteration.
- Backlog Refinement: The product backlog is constantly updated with new tasks, bug fixes, and enhancement ideas based on user feedback or new insights.
- Ongoing Improvement: Based on releases and feedback, the team can adjust priorities and refine features in upcoming iterations.

5. Project Schedule

6. References

- [1] Google Workspace. Retrieved from <https://workspace.google.com/products/drive/>
- [2] Microsoft SharePoint. Retrieved from <https://www.microsoft.com/en-us/microsoft-365/sharepoint/collaboration>
- [3] Dropbox. Retrieved from <https://www.dropbox.com/business>

Appendices